

KV Admission: Learning What to Write for Efficient Long-Context LLM Inference



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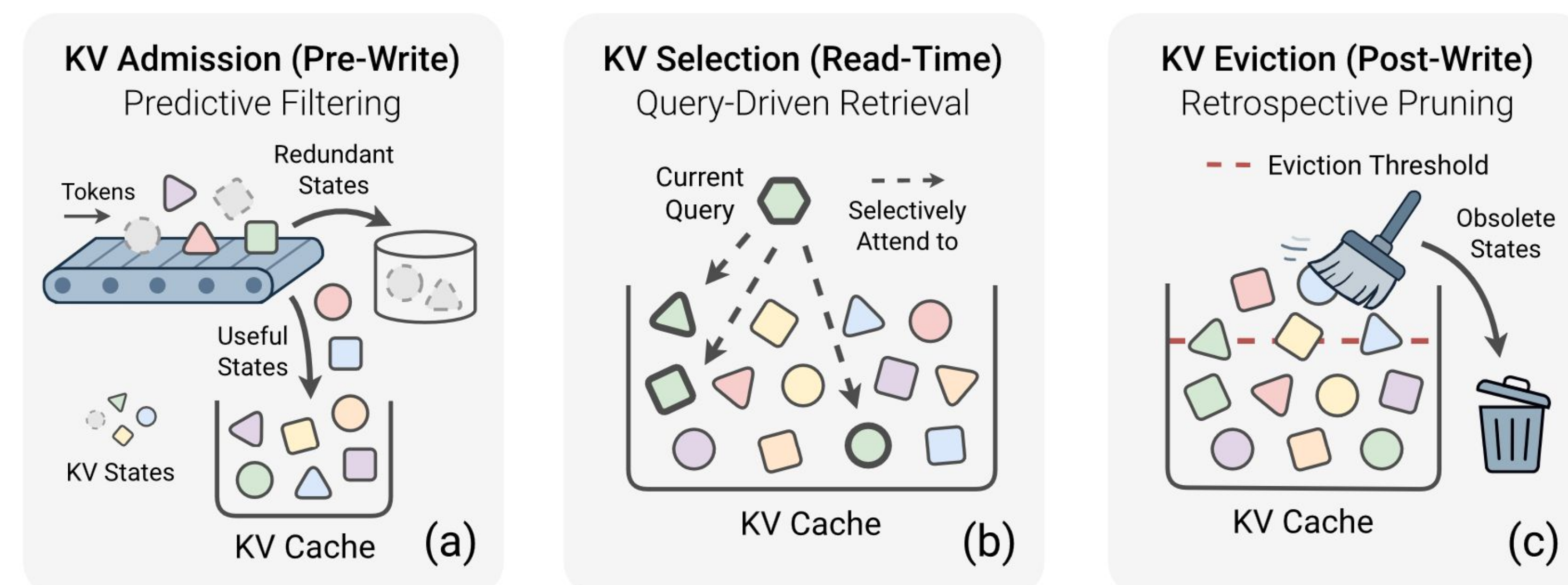
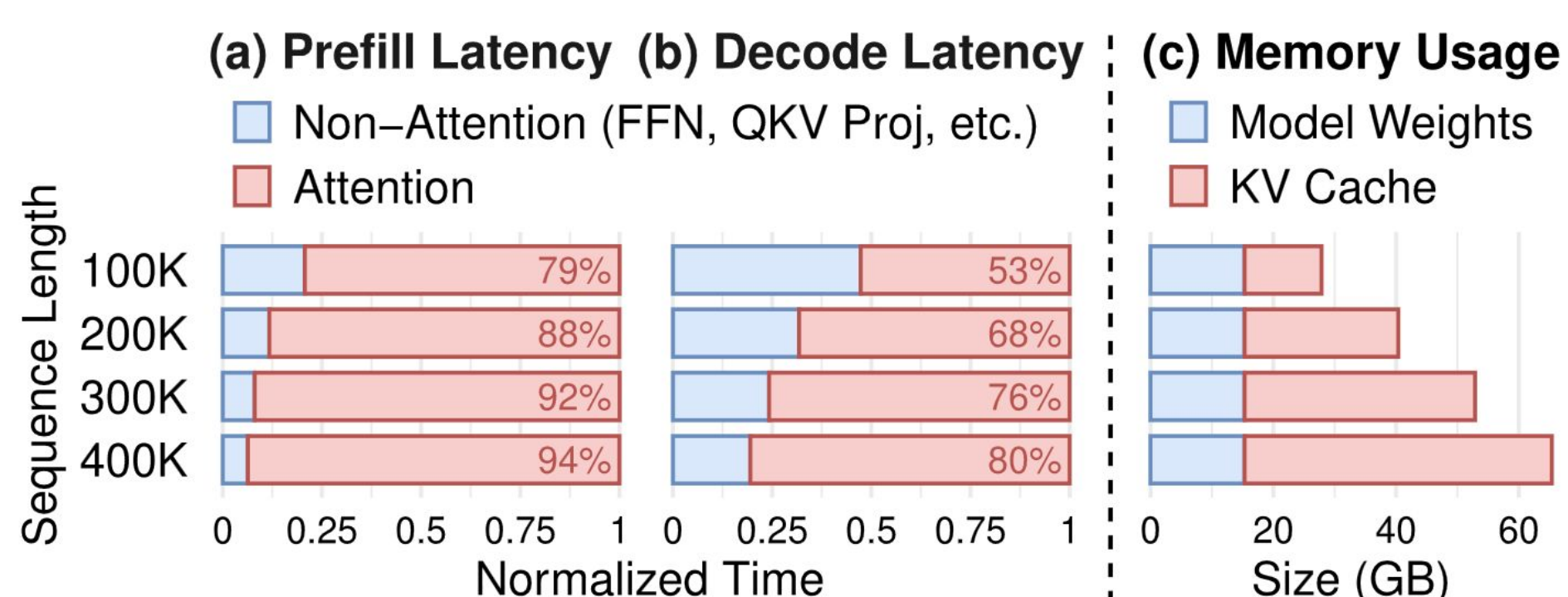
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TL;DR Not every token is worth writing to the KV cache — *learn* which ones are.

1. The missing primitive: KV Admission

Long-context inference is bottlenecked by attention: at 400K tokens it takes **94% of prefill** and **80% of decode** time on Llama-3.1-8B, while the KV cache grows linearly with context. Yet existing methods act only after a token's KV is written — so every token is *admitted indiscriminately*.

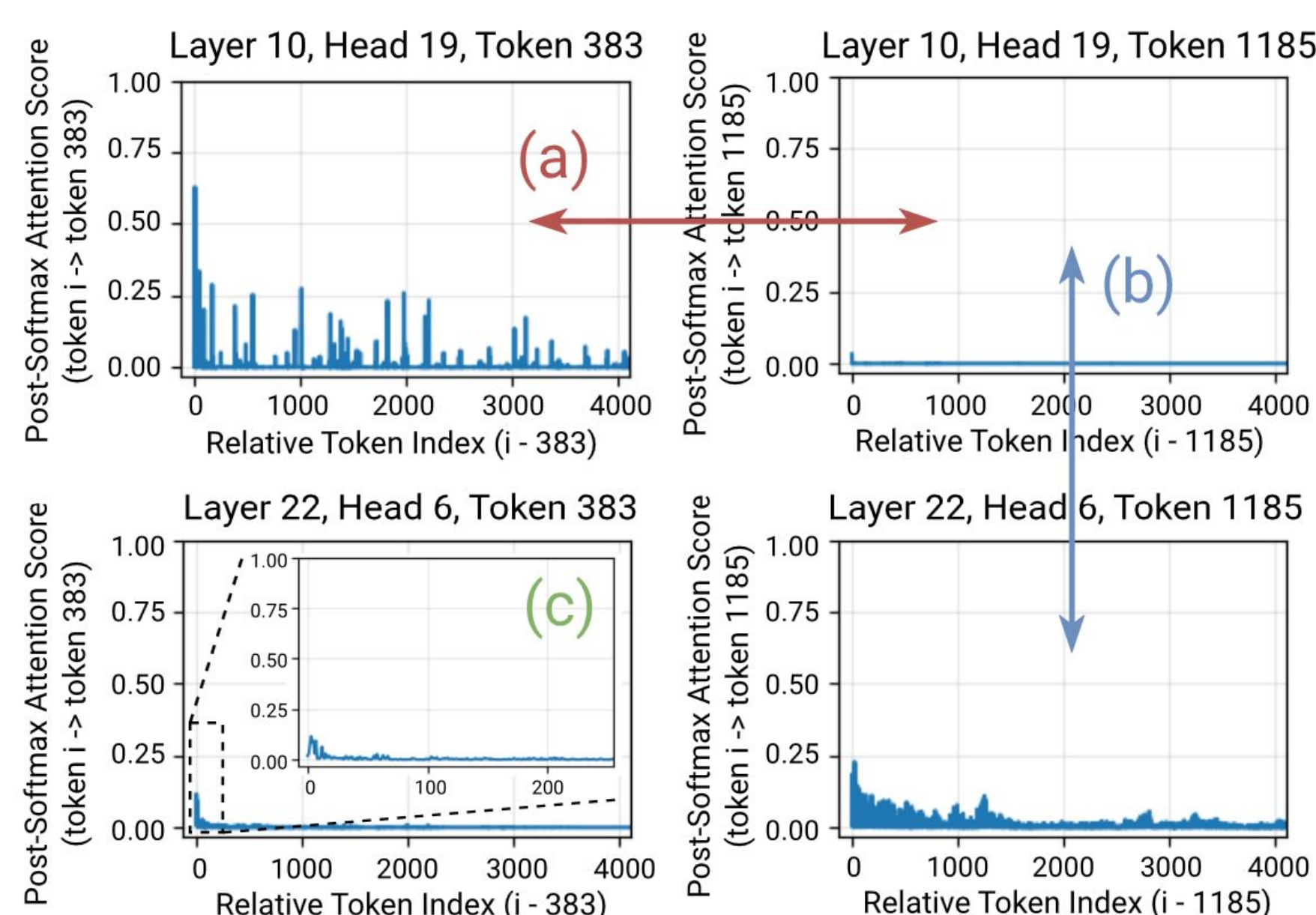


Our work

Prior work

Selection **trims what is read**; Eviction **retires history** that has served its purpose. Admission adds the missing step: deciding **what to write** in the first place. Together they span a token's lifecycle — write, read, retire.

2. What attention patterns tell us

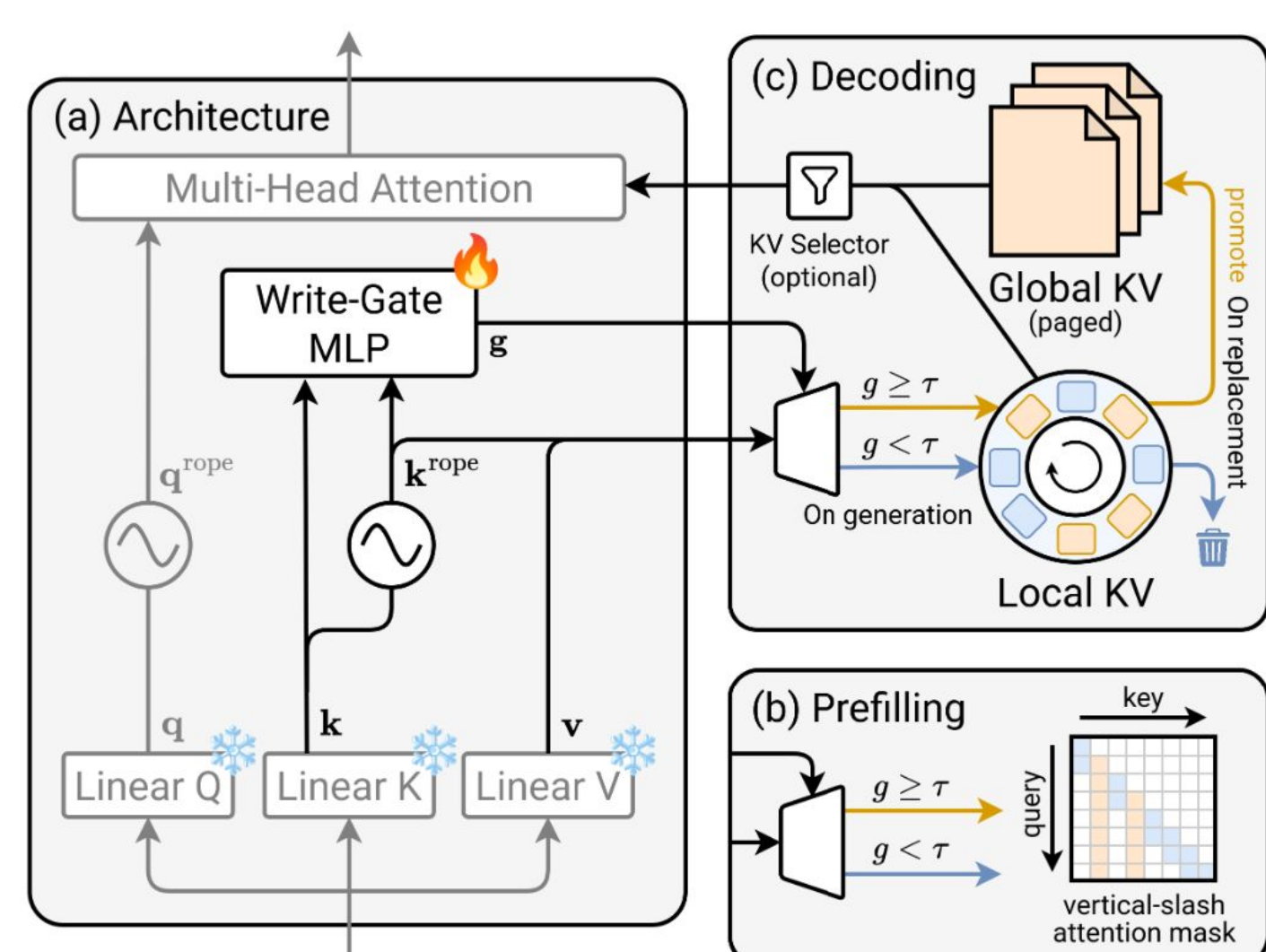


(a) **Skewed utility**
A few tokens draw most attention → filter the rest

(b) **Head-specific**
Heads value different tokens → decide per KV head

(c) **Local recency**
Recent tokens matter briefly → keep them for a while

3. Write-Gated KV (WG-KV)



- A 2-layer **Write-Gate MLP** per KV head scores every token: $g \in [0, 1]$
- $g \geq \tau \rightarrow$ **global cache**, kept long-term
- $g < \tau \rightarrow$ **local cache**, live a sliding window, then discarded
- Backbone frozen; only the gate is trained ($\approx 0.4\%$ extra params)

- Admission is a **hard 0/1 decision** at inference — hence non-differentiable
- We relax it into a **continuous re-weighting of attention**, so gradients can flow

$$\text{Attn}(q_i, K_{\leq i}, V_{\leq i}) = \sum_{j \leq i} \frac{\exp(q_i \cdot k_j / \sqrt{d})}{\sum_{p \leq i} \exp(q_i \cdot k_p / \sqrt{d})} v_j \quad \dots \text{Standard Attention}$$

$$\widetilde{\text{Attn}}(q_i, K_{\leq i}, V_{\leq i}) = \sum_{j \leq i} \frac{\exp(q_i \cdot k_j / \sqrt{d}) \cdot m_{ij}}{\sum_{p \leq i} \exp(q_i \cdot k_p / \sqrt{d}) \cdot m_{ip}} v_j \quad \dots \text{Write-Gated Attention}$$

$$m_{ij} = \max(\mathbb{1}(i - j < W_{\text{local}}), g_j) \quad \dots \text{Soft Admission Mask}$$

- Fused attention kernels (FlashAttention) are optimized only for $\exp(q \cdot k)$. To stay kernel-compatible, We fold the gate into an **additive log-space bias**:

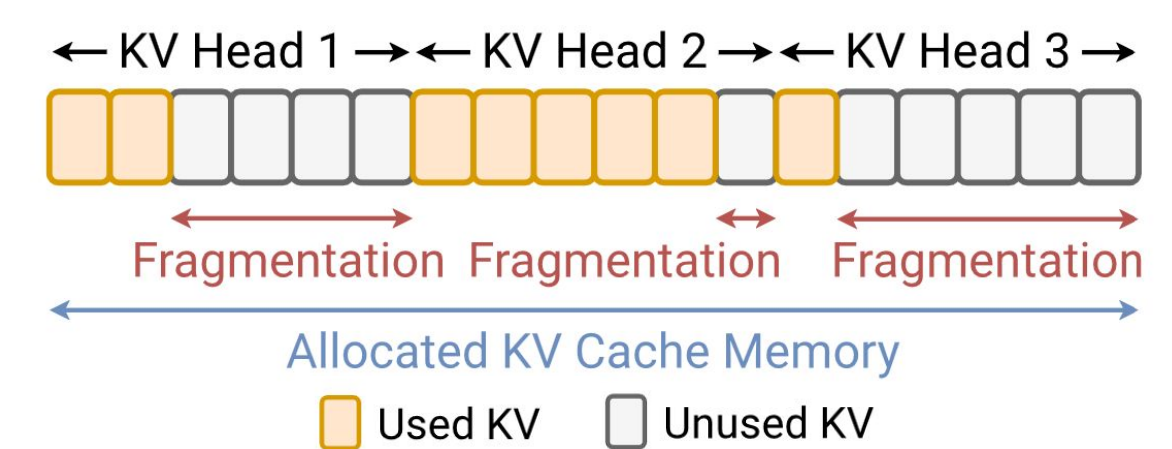
$$\exp(q_i \cdot k_j / \sqrt{d}) \cdot m_{ij} = \exp(q_i \cdot k_j / \sqrt{d} + \log(m_{ij})) \Rightarrow \widetilde{\text{Attn}}(q_i, K_{\leq i}, V_{\leq i}) = \text{Softmax}(q_i K_{\leq i}^T / \sqrt{d} + B_{\text{gate}}) V_{\leq i}$$

- Training objective**: last-layer MSE compared to the original full-attention model

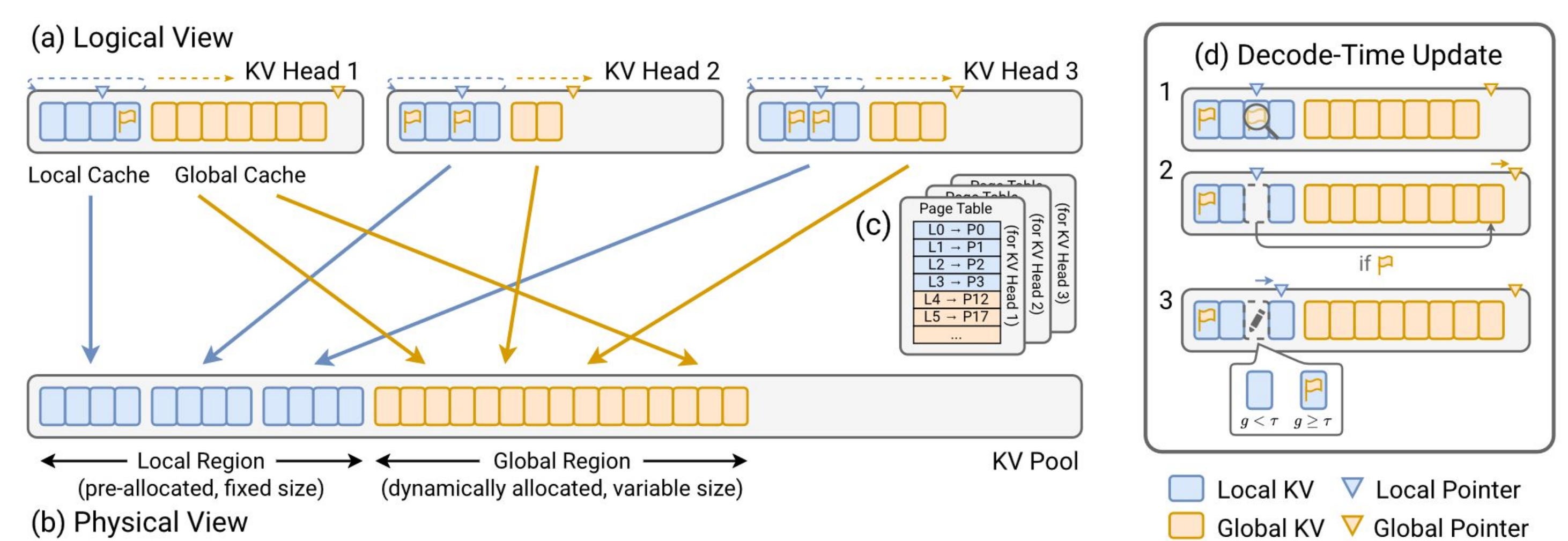
$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{distill}} + \lambda \mathcal{L}_{\text{sparsity}} \quad \mathcal{L}_{\text{sparsity}} = \frac{1}{L \cdot H \cdot T} \sum_{l, h, t} (g_{l, h, t} + g_{l, h, t} (1 - g_{l, h, t}))$$

4. System design: turning sparsity into speedups

Per-head decisions make the cache *ragged*: contiguous buffers would fragment memory or force costly reallocation. Three pieces turn sparsity into speedups:



- Paged dual cache**. Per-head page tables map **local** and **global** caches onto one shared KV pool.
- Prefill**. Vertical-slash sparse attention skips non-admitted tokens instead of dense $O(N^2)$ attention.
- Decode**. The token leaving the local ring buffer is promoted if $g \geq \tau$, else dropped. Attention runs on variable-length paged kernels.



5. What does the gate learn?

Content head

keep proper nouns and key entities (Layer 25, Head 1)

Nikolai Medtner was born in Moscow on 24 December 1879, according to the Julian calendar, or 5 January 1880 by the Gregorian calendar. He was the son of Karl Petrovich Medtner (1846–1921) and Alexandra Karlovna Goedicke (1843–1918), and the fifth of their six children. Both of them were of German origin.

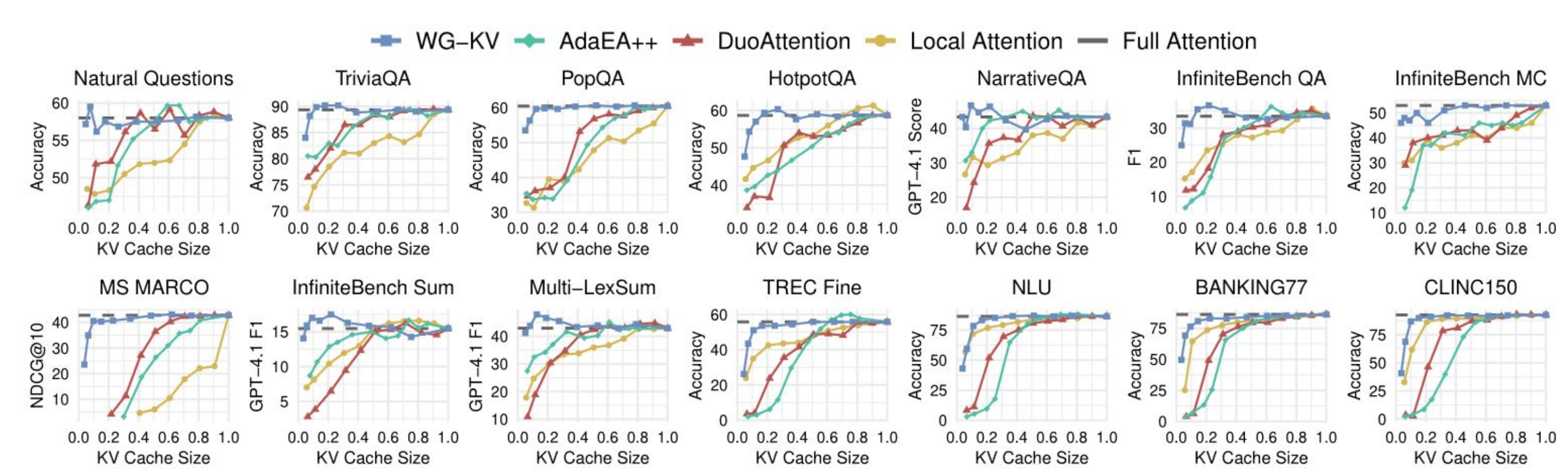
Numerical head

keeps dates and numerical values (Layer 29, Head 8)

Nikolai Medtner was born in Moscow on 24 December 1879, according to the Julian calendar, or 5 January 1880 by the Gregorian calendar. He was the son of Karl Petrovich Medtner (1846–1921) and Alexandra Karlovna Goedicke (1843–1918), and the fifth of their six children. Both of them were of German origin.

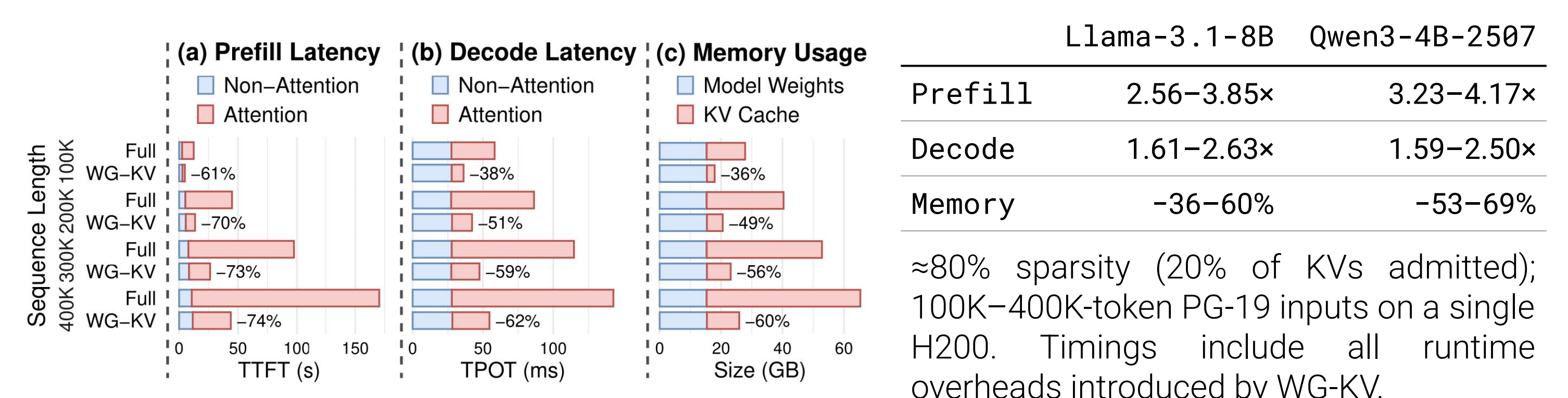
No rules given: each head keeps little, yet together they cover the entire passage.

6. Accuracy holds as the budget shrinks



- Across HELMET tasks (RAG, long-doc QA, re-ranking, summarization, many-shot ICL), WG-KV beats static and training-free admission baselines.
- On information-dense tasks, **near-lossless accuracy even at 90% sparsity**.

7. End-to-end speed and memory savings



	Llama-3.1-8B	Qwen3-4B-2507
Prefill	2.56–3.85×	3.23–4.17×
Decode	1.61–2.63×	1.59–2.50×
Memory	–36–60%	–53–69%

$\approx 80\%$ sparsity (20% of KVs admitted); 100K–400K-token PG-19 inputs on a single H200. Timings include all runtime overheads introduced by WG-KV.

8. Composes with KV Selection and Eviction

w/ Selection –76% KV memory, –65% KV reads and –65% attention latency per decode step, with nearly identical accuracy

w/ Eviction Qwen3-4B reasoning under a strict memory cap: eviction alone 26.7%, admission alone 66.7%, both 80.0% — matching the unbounded cache.

